THE UNITED REPUBLIC OF TANZANIA

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA PRIVATE SECONDARY SCHOOL ASSOCIATION (KIPSSA)

FORM TWO PRE-MOCK EXAMINATION

041 BASIC MATHEMATICS

Time: 2½ Hours **May 2025**

INSTRUCTIONS

1. This paper consists of a total of 10 questions.
2. Attempt all questions.
3. Each question carries 10 marks.
4. Mathematical tables and non-programmable calculators are not allowed
5. Write your examination number on every page of your answer sheets

FOR EXAMINER'S USE ONLY		
QUESTION NUMBER	**SCORE**	**EXAMINER'S INITIALS**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

1. (a) Three bells are set to ring at intervals of 15 minutes, 20 minutes, and 24 minutes. If they started ringing all together at 8 a.m., find the time the three bells will ring together for the second time.

(b) Seypalm uses $\frac{1}{3}$ of his money to buy books, $\frac{1}{4}$ of his money to buy pencils and a mathematical set, and he remains with 35 000 shillings.

(i) How much money did he have at the beginning?

(ii) How much money did he use to buy books?

2. **(**a) Evaluate $\frac{2}{25} \times 0.737$ to:

(i) One significant figure

(ii) Three decimal places

(b) Seyalm bought 50 bottles of milk for **70** children. If the capacity of each bottle is 300 ml, find the amount of milk in litres that Palm bought.

(c) A lorry carries 7.2 tonnes of sand from mining area to the site, on the way, 250 kg of sand falls off or is blown away. What is the mass of sand remained at the end of the journey? (Give the answer in tonnes)

3. (a) The perimeter of isosceles triangle is 15 cm. If the base is 7 cm long, represent this information in a diagram, and then find the lengths of the remaining sides.

(b) if $2x + 70$ and $3x + 20$ are supplementary angles, determine the value of x

(c) The interior angle of a regular polygon is 156° greater than the exterior angle. How many sides does the polygon have?

4. (a) Solve the equation: $3x^2 - 7x - 6 = 0$ by completing the square.

(b) Solve the following simultaneous equations by the method of substitution:

$$\begin{cases} 2x + y = 10 \\ 3x - 2y = 1 \end{cases}$$

5. (a) Given $a: b$ = 5:2 and $b: c$ = 3:4, find the value of:

(i) a:c

(ii) $a: b: c$

(b) A car, which its buying price is 12 500 000 shillings was sold at the profit of 15%. Find the profit made and the selling price.

(c) Calculate the simple interest on 300 000 shillings deposited in a bank that offers an annual rate of 6% simple interest for three years.

6. (a) Find the image of B (3, 4) after a reflection in the line $y = -x$ followed by another reflection in the line $y = 0$.

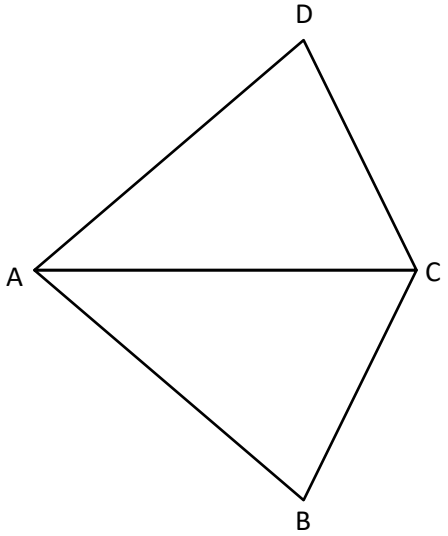
(b) Find the equation of a straight line which passes through $(\frac{-1}{2}, 5)$ and whose x −intercept is$\frac{-3}{4}$

7. (a) Simplify and rationalize the denominator of the following expression:

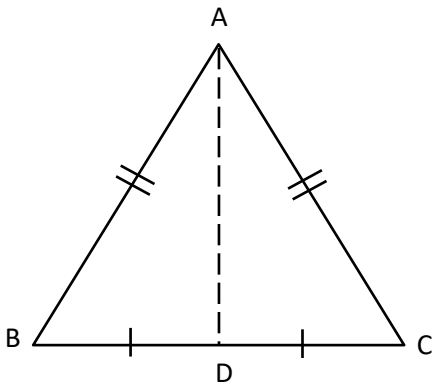
$$\frac{1}{5-3\sqrt{2}} - \frac{1}{3-2\sqrt{2}}$$

(b) Find an expression for x in terms of y; log $(\frac{x}{y})$ - log $(\frac{y}{x})$ = 1

8. (a) In the following figure, $\overline{AC}$ bisects $\angle BAD \; and \; \angle BCD$, prove that $\overline{AB} = \overline{AD}$



(b) Triangle ABC is an isosceles triangle in which $\overline{AB}$ *and* $\overline{AC}$ are equal. If D is the midpoint of $\overline{BC}$, prove that $\Delta ABD \cong \Delta ACD$



9. (a) A rectangle measures 15 cm by 11 cm. What is the length of the diagonal?

(b) The angle of elevation of the top of a building 24 m high is observed from the top and from the bottom of a vertical ladder, and found to be 45° *and* 60°, respectively. Find the height of the ladder.

10. (a) Given that U= $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $A = \{1, 3, 5, 7, 9\}$

(i) How many subsets are in A'?

(ii) iii Find: A'$\cap B$ if B = $\{1, 2, 3, 4\}$

(b) A Tailors' Association surveyed the number of sewing machines that each primary association had and recorded as follows:

3	8	5	4	5	5	6	7	7	4	6	6
7	4	4	6	7	5	4	3	7	5	7	8

3.

(i) Construct a frequency distribution table.

(ii) Construct both frequency polygon and Histogram on the same graph.